i. CHARACTERISTICS OF EACH COMPONENT ON THE ELECTRICITY' D}DEMONSTRATION BOARD BOARD

We recommend that you have a "play" with the demonstration board before you read all of this information, to give you some familiarity with how some of the components behave. Circuits you can try are on an attached sheet.

Each component (buzzer, motor, etc.) on the board will work when DIRECT CURRENT (from a battery or D.C. power supply) is connected to them. They are not designed to work with alternating current (A.C.), which is the type of current we receive through the mains power supply. Each component has a certain amount of Resistance to electric current. It is helpful to be aware of the approximate resistance of each component, especially when you want to predict what will happen before you connect up a circuit.

The resistances liven are the MEASURED RESISTANCES, when a 6 volt battery was connected to them. An explanation of resistance is given in the section on "Electricity - the main concepts"

(a) Globes - There are two types, the only difference being their resistance. (i) one marked 6v 2w - resistance 27 ohms. (ii) the other marked 3.8 V 0.3 A - resistance 18 ohms. Globe (ii) glows more brightly than globe (i) when each is separately connected to a 6 volt supply. (Try it).

The electric current flows into the base of the globe, through the filament and out of the metal side of the globe. The current can also flow in the reverse direction. The reason for the glass casing is to protect the filament from being knocked or touched when it gets
white hot. and to keep oxygen away from the filament. If the filament was exposed to oxygen when it was hot it hot, it would instantly burn out.

(b) continuity tester - This has many uses. When the two probes are connected together (that is, there is no resistance between them) the light emitting diode (L.E.D.) lights up. Therefore, if the L.E.D. lights up brightly when the probes are being used, the ~two points being touched by the probes are connected electrically. Other uses are listed in the ex]experiment section. The continuity tester works best when connected to a nine volt D.C. supply.

(c) variable resistor - (Also known as a Potentiometer, or POT for short). The resistance between the centre socket and the ~o outer sockets can be changed from 10 ohms up to 10k ohms (10,000). The resistance between the two outer sockets is always 10k ohms. There is a protective 10 ohm resistor on the central socket of the pot on the demonstration board. If there was no resistor, the pot could BURN OUT Ok~ if a power supply is connected only across the pot. This is a danger with the students' pot, as there is no resistor attached to the central terminal. Students should always ensure that the pot is in series with another component such as a globe.

(d) motor - The resistance of the motor is around 18 ohms when running with a 6 volt battery supply. The motor will change direction of spin if the leads in the two motor sockets are swapped. The optimum voltage supply to run the motor is only 3 volts, but it can run comfortably at 9 volts, provided it is not left running for a long period of time. It will spin at 6 900 revolutions per minute at these voltages, if not restricted.
Provided in the kit is one means of restriction - an icypole stick, with a hole to fit onto the motor shaft. See later experiments with the motor for details.

(e) buzzer - The resistance of the buzzer is around 600 ohms. There are two other sockets connected to the POSITIVE TERMINAL of the buzzer, each with an additional l50 ohm resistance. These resistors decrease the current going into the buzzer, and hence make in softer. If the students are using the demonstration board, you may PREFER to have them use only the "soft" volume socket. You can do the same with the students boards (as outlined in the later experiments). The buzzer needs at least S volts power supply with the positive of the supply connected to the positive of the buzzer. the following information provides suggestions for extended experimental options in the physics using the Electricity Demonstration Board. I recommend that you put variouis problrems to the students that requires them to come up with some of the methods ~(or better ones) suggested below~ for determining the characteristics of the solar panels ,and motor. In other words. let them solve the problem.

A.CHARACTERISTICS OF SOLAR PANELS.

There are various aspects of the solar panels that can be investigated.

(a) Plotting the light intensity on the panels against the current and voltage output.
The output of the panels can be placed in series with a milliammeter and 100 ohm resistor, with a voltme[er in parallel ~ith the 100 ohm resistor. A light intensity meter used b>photographers can be used to measure the light intensity at the panel surface usin~ a 150 ~,~tt ii~ht globe as the source.

(b) Using the set up as in (a), determine the percentage efficiency of the solar panels. For a particular distance of the panels from the ligh~ source (say x cm), the surface area of a sphere of radius x can be calculated (say area Y cmz). Assuming the light globe actually emits 150 joules per second as light, and that the light is evenly distributed over the imaginery surface of the sphere or radius x, it is easy to calculate the amount of light energy per second reaching the solar collectors. Using_' the P = IV for the output of the solar panels, you can determine the percentage efficiency of the panels. This efficiency can be determined for various distances from the globe.

~c) A variation on (b) could be to use colour filters to see if the panels are sensitive to particular colours. Ad.~ustment of distance may be necessary to keep the overall light intensity on the panels constant.

There are a range of extensions to this topic, including visiting showroom of the latest solar technology, and writing up on their research. Students could also construct their o~ solar water heater, and determine the efficienc~~ of energ~ transfer.

T]~_ ELEtmfRIC ~FoTOR

The measureable parameters relevant to the motor are (a) its ms~ximum speed of rotation, (b) the best operating voltage, and the current dra~ by the motor at this voltage (the power consumed for various voltages can be determined); (c) the output power of the motor, and hence its relative efficiency' for a Oarticular load on the motor: (di the ms~ximum torque of the motor. Some of these parameters are more difficult to measure than the others, particularly (c) and (d).

SUGGESTIONS

(a) Use a stroboscope to determine the frequency of rotation for various inpu~ voltages immx. 9 volts, for short time only). Plot f vs input voltage.

ib) From the graph of (a), determine the minimum voltage for ms~ximum f.

(c) Put a piece of pencil with its lead removed on the end of the shaft so that it can act as a pulley. A long cotton thread with a light weight can be attached to the pulley. If the motor is then raised to a height of at least 2 metres, the students can then attempt to determine the power output (rate of change of potential energy of the weight) versus the input power to the motor. Because of the speed of the motor, students may wish to gear the speed of rotation do~, or file most of the pencil shaft down - the motor can make a light weight move ~.~]ERY rapidly. The challenge is to get the motor operating_ at its optimum efficiency!

ld) By having a long ic.vpole stick firmly attached to the shaft lyou may want to file a

The school will probably have other motors that students can compare. Students may not be aware of how many motors are in the average house. There maybe anything up to forty motors in one home, including in toys (trains and slot cars), appliances with moving parts - vacuum, dishwasher, washing machine, video, tape recorder and player, record player, exhaust fans, hair dryer, blow heaters, cooling fans, electronic type~Titers and printers, food processors. mix-masters, electric can openers, tumble driers, refrigerators and freezers (you hear the motor going on or off occasionally), electric tools in the workshop - drills, grinders, electric planes and touters, electric saws, etc. Try and estimate how many motors you have in your home.

7. ENERGY CHANGES - THE ~I4iN CONCEPTS

There are a number of energy transformations that can be sho~m using the demonstration board.
(a) If you have a circuit with a battery and a light globe {as with a torch), the energy changes that take place are - chemical energy (stored in the battery), electrical energy (transferred through the wires by means of electric current) and finally light and heat energy (given off by the light globe).
(b) The motor produces mechanical energy (working energy) ~'hich can be used to produce kinetic energy (energy,' of motion).
(c) The buzzer produces sound energy.

It is obviously not important that students know and understand all these names of different types of energy, but the concept of energy TRANSFER, starting in the battery, passing through the wires to the different components (or appliances) and then being released in a different form is important.